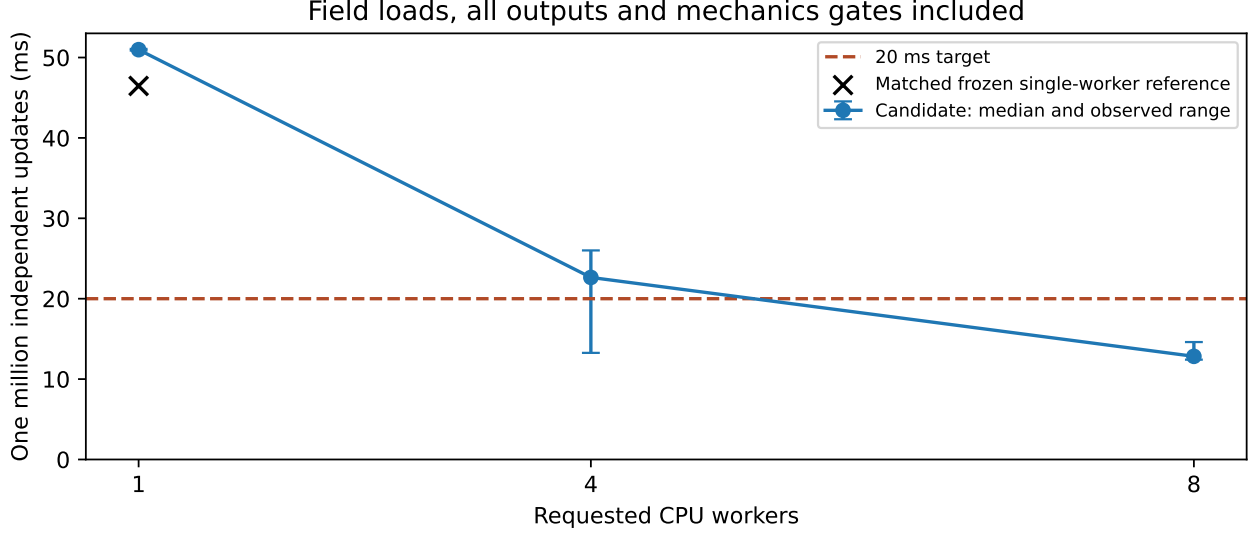


Supported-contact compiler integration package

Verified native interface, portable algorithm and explicit limits
7 October 2026

Result. A versioned C-compatible batch boundary and indexed implementation are prepared for a scoped BKF / Vektor Flow import. One million independent supported-contact responses take **12.830 ms median with eight workers**, range 12.417–14.606 ms. Every one of seven samples meets the 20 ms target. 54 focused tests pass, including native/reference, worker determinism, wide-size controls and a C11 header probe. Compiler repositories are unchanged.



Requested workers	Median ms	Observed range ms
1	50.975	50.863–51.071
4	22.646	13.261–26.005
8	12.830	12.417–14.606

Fair comparison. Matched-layout single-worker reference: 46.463 ms; candidate: 50.975 ms. There is no demonstrated single-worker algorithmic speedup. Four workers miss the controlled median target. Earlier 81.4 ms used a different harness and is not the paired reference. All attempts remain archived.

Measured scope. Field-major loads, all eleven stores, worker entry and finite/branch/energy gates are included. Preparation, input validation, allocation, detection, changing loads/frames, impacts, group solving and contact-history updates are excluded. Repeated independent fixtures are not a million interacting bodies. No all-case 2× gate, compiler/GPU acceptance or experimental gain is claimed. The measured host is AMD Ryzen 7 3700X, Linux x86-64; the C++17 build uses OpenMP, `-O3 -Wall -Wextra -Werror`, without fast-math.

Algorithmic correctness before compiler tuning

The supported sphere/disk has scalar central inertia I , radius R , translation v , rolling speed ω , force f and independent rolling moment M_r . Use rim speed $q = R\omega$ and force-equivalent couple $b = M_r/R$:

$$\begin{bmatrix} \dot{u} \\ \dot{q} \end{bmatrix} = \begin{bmatrix} F_d/m \\ 0 \end{bmatrix} + \begin{bmatrix} 1/m + R^2/I & -R^2/I \\ -R^2/I & R^2/I \end{bmatrix} \begin{bmatrix} f \\ b \end{bmatrix}, \quad u = v - q.$$

Both mobility rows now have consistent linear units. Reciprocals and capacities are computed once per update. Continued nonzero sliding/rolling has known signs and skips active-set enumeration. Zero motion uses explicit static/onset candidates; events stop at slip/rolling arrest before constraints are recomputed.

Fixed defects. The old branch tolerances mixed linear and angular units: a valid frictionless 10^{-8} N drive could be rejected for small bodies. The corrected Python/native paths work for radii 10^{-12} – 10^6 m. Axial spin is set to exact zero at its impulse bound, preventing roundoff reversal. Nonfinite energy/response states are rejected, including static impulse overflow without work. No material value or restitution coefficient is fitted or changed.

Impulse identity. Independent angular impulse remains separate from the linear-impulse lever moment. The scalar branch satisfies

$$\Delta \mathbf{L} = \mathbf{r} \wedge \delta \mathbf{p} + \delta \mathbf{L}.$$

The output rolling moment channel is a physical independent impulse, not a replacement for the lever term or a new definition of the full spin direction. Native scalar outputs are signed fixed-axis integrals. They are not automatically coefficients on a changing motion-defined t/s basis: apply each branch direction before accumulating vectors, rather than multiplying by an endpoint direction. The user's full contact-relative $\hat{\mathbf{t}}$ and full angular $\hat{\mathbf{s}}$ remain binding in allowed spatial embeddings. Their general static/partial-arrest closure is still unresolved.

Verification. 400 updated controls pass at maximum scaled native/Python error $1.04 \cdot 10^{-15}$. Another 2,000 controls span masses 10^{-6} – 10^6 kg and radii 10^{-9} – 10^6 m with maximum scaled error $1.02 \cdot 10^{-15}$. Worker counts produce bit-identical responses, and every timed million-response output is compared against the single-worker output. Energy, composition and 100 rotated supported controls remain passing. An initial C-probe test used noncontiguous storage for a field-major pointer; its failure is retained and the corrected probe passes the unchanged native contract.

Experimental local memory. A separate one-to-three-mode spring/slider component uses midpoint elastic storage and a static test followed by a dynamic convex return map. Small factorizations are cached; at most 27 active sets are needed. 300 randomized coupled-mode controls check passivity and permutation invariance. Opening applies no impulse and transfers stored energy to an explicit internal-mode ledger. The caller must retain or physically relax that energy; it is not silently deleted or assumed to be measured heat. This component is not called by the native batch and is not a completed impact-deformation model.

Compiler import boundary and readiness

`supported_backend/batch.h` is C11/C++17 compatible. ABI v1 exposes version/field counts, host worker limit, input validation and synchronous update. The machine-readable field names and units are in `schema.json`; the full ownership and lowering contract is in `INTEGRATION.md`.

Use field-major IEEE Float64 arrays and one body index k :

`input[field*count+k],      output[field*count+k].`

There are fifteen inputs and eleven outputs. The caller owns disjoint buffers; no pointer is retained and no per-body heap allocation occurs. Explicit workers operate on independent responses. Success is `SIZE.MAX`; global invalid arguments have a distinct status, otherwise return the earliest failed body. Discard the entire output after failure. No fallback law is substituted.

Python's `PreparedBatch` owns an immutable validated input copy and can write directly into caller-provided output storage. Preparation and allocation must be measured separately for a full application. Native callers validate when inputs change and honor the buffer contract. The synchronous update checks branch, finite-range and energy/work admissibility.

Ready for scoped import: native supported branch, field/units schema, failure semantics, independent body loop, scalar and wide-scale importer oracles, ownership contract, warning-free source and explicit benchmark evidence.

Not qualified: general collision/contact groups, arbitrary tensor inertia inside this scalar branch, evolving full t/s history, a physical coupled patch capacity law, independent empirical calibration, native/WASM/GPU compiler equivalence or main-compiler acceptance. The new memory component stays experimental. Measured inertia input elsewhere in the 3D engine remains separate. Normal/tangential restitution belongs to impact branches and is not reapplied to this sustained-contact law.

Import procedure. Start from the compiler's accepted paired checkpoint and current Section 0 authority. Add new fixtures without editing old oracles. Compare all eleven channels with the archived physical scales; run actual target, ownership and preservation checks before claiming a compiler integration. This package changes no compiler seed/main/baseline or hardware acceptance.

Experimental accuracy. Previous glass/public-data residuals are unchanged. No new measured improvement is shown. Tangential history/partial slip is established prior physics, e.g. [Maw-Barber-Fawcett \(1976\)](#) and [LAMMPS granular documentation](#); these motivate local state but do not validate this reduced component or supply every independent material parameter.

Reproduction. Package README and `supported_backend/INTEGRATION.md` give build/run commands. Benchmark source/receipts are in `research/supported-batch-optimization/`; `run-v1/v2/v3` retain all attempts. The downloadable bundle includes the native core, Python reference, focused tests and import oracles. Whole-engine regressions require the full repository. Source/member hashes and extracted-bundle test results accompany the final artifact receipt.